Cybersecurity Basics 🔐

Cadette Cybersecurity Badge — Girls Build Bots

Presented by your robotics mentors

Knowledge over fear

Today you get the honest, technical version:

- How the internet **actually** moves your data
- What encryption **really** does
- What happens to information you hand over

Someone who understands the system protects herself better than someone who's just been warned about it.

The Packet Game 📨

Real messages don’t travel whole.

They’re split into **packets** that hop router to router and reassemble at the destination.

1. Split your message across **three numbered cards**
2. Hand them to different **routers**
3. They arrive out of order — one gets delayed
4. The receiver **reassembles by number**

Round 2: the snooping router 👀

An unprotected card gets **read aloud**.
Debrief: your data crosses machines you don’t control.

Which is exactly why encryption exists.

Encryption, for real 🔑

Encrypt with the cipher wheel. Send it through the snooper.
The snooper now gets gibberish.

From cipher wheel to HTTPS 🔒

Modern encryption is this same concept — with math too hard to break.

HTTPS — the padlock in your browser — means it's happening automatically.

New rule of thumb: no padlock, no passwords.

Where does your data go? 🗺️

For each scenario card, map it out:

- What **data** is generated?
- **Who** receives it?
- Where might it **travel from there**? *(the app's server, advertisers, "free" quiz companies)*

Metadata 

Even **without the content**, the *when / where / who* of your activity tells a story.

Data is the price of “free.”

You should know what you’re paying.

NOVA Cyber Lab round 💻

Play the defense-focused lab game.

Connect what’s on screen back to the cipher wheels and the packet cards.

Robot moment 

Our robot and driver station exchange **packets** over a locked-down field network.

The connection is secured so **only our station** can command our robot.

Closing 🎯

Name: 1. One thing you understand about the internet that you didn’t this morning 2. One **behavior you’ll change** because of it

What you accomplished ✅

- Explained how data travels the internet in packets through routers
- Encrypted and decrypted a message, and explained what HTTPS signals
- Mapped where personal data goes after an everyday online action
- Explained what metadata reveals even without message content

Level up 🚀

- **pbs.org/wgbh/nova/labs/lab/cyber/** — the full NOVA Cybersecurity Lab
- **cyber.org** — free cybersecurity coursework
- Check the padlock before you type a password. Every time.

📣 Share Your Impact

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This is how the program grows — and how the next troop and team get better materials.